Getting Started: UN World Population Prospects 2024 Dataset

This guide covers the three files provided for the 2026 DataQuest competition, what's in them, and how to load and start working with them.

1. What's in the box

File	Contents	Rows	Years covered
WPP2024_Demographic_Indicators_Medium_csv.gz	The main dataset: the UN's single best-estimate ("Medium") series — historical estimates plus the medium-variant projection, as one continuous series	84,360	1950–2101
WPP2024_Demographic_Indicators_OtherVariants_csv.gz	**Optional.** Alternative projection scenarios (high/low fertility, constant mortality, zero migration, confidence intervals, etc.) — projected years only	662,919	2024–2101
WPP2024_Demographic_Indicators_notes.csv	The indicator codebook — what each column abbreviation means and its unit of measurement	59	—

Both .gz files are gzip-compressed CSVs. Programming tools (pandas, R's read_csv) can read them directly without a separate unzip step (see Section 3) — but **Excel cannot open a .gz file directly**. If you want to use Excel:

- Extract the file first (double-click on Mac; on Windows, a tool like 7-Zip).
- Check the extracted file's name — the UN's archive stores it without a .csv extension (it ends in _csv, not .csv), so most extraction tools will reproduce that extension-less name. Rename it to end in .csv before opening, or Excel won't recognize it as a spreadsheet.
- Open the renamed file in Excel as usual.

Start with the Medium file. It's the core dataset for most analyses — one row per location per year, with population, fertility, mortality, and migration indicators as columns. Only bring in OtherVariants once you want to ask questions about projection uncertainty (see Section 5).

2. Key columns to understand before doing anything else

Column	What it is	Why it matters
Location	Name of the country, region, or grouping	—
ISO3_code	3-letter country code (e.g. USA, KEN)	**Only populated for actual countries** — blank for regions/aggregates. Useful filter.
LocTypeName	Indicates what kind of row this is	Some rows are individual countries; others are regional, income-based, or organizational groupings. Not all of these

Column	What it is	Why it matters
		are labeled consistently — see the pitfalls note below.
Variant	Which projection scenario the row belongs to	In the Medium file, this is always "Medium". In OtherVariants, it takes 18 different values (see Section 5).
Time	Calendar year	Ranges 1950–2101

The location field in this file doesn't only contain individual countries. It also contains regional aggregates, income-group aggregates, and a long tail (~230+) of ad hoc groupings. If you don't work out how to separate actual countries from these groupings before doing any cross-country comparison or aggregation, you will end up double-counting population.

3. Loading the data

Python:

```
import pandas as pd

medium = pd.read_csv(
   "WPP2024_Demographic_Indicators_Medium_csv.gz",
   compression="gzip",
   encoding="utf-8-sig",  # avoids a garbled first column name
   low_memory=False       # avoids a noisy (harmless) mixed-type warning
)
```

R:

```
medium <- read.csv(
  "WPP2024_Demographic_Indicators_Medium_csv.gz",
  fileEncoding = "UTF-8-BOM"
)
```

4. Indicator reference (from the notes file)

Every column in the Medium and OtherVariants files is one of the indicators below, grouped by topic. Use this as your lookup table rather than guessing from the abbreviation. Note: where a male/female or age-specific breakdown shares the same underlying concept and unit (e.g. Deaths / DeathsMale / DeathsFemale), rows are combined below for readability — the combined description is a summary, not a verbatim quote of each individual entry. For exact per-column wording, see WPP2024_Demographic_Indicators_notes.csv directly.

Population

Code	Name	Unit
TPopulation1Jan	Total Population, as of 1 January	thousands
TPopulation1July	Total Population, as of 1 July	thousands
TPopulationMale1July	Male Population, as of 1 July	thousands
TPopulationFemale1July	Female Population, as of 1 July	thousands
PopDensity	Population Density, as of 1 July	persons per square km

Code	Name	Unit
PopSexRatio	Population Sex Ratio, as of 1 July	males per 100 females
MedianAgePop	Median Age, as of 1 July	years
NatChange	Natural Change (Births minus Deaths)	thousands
NatChangeRT	Rate of Natural Change	per 1,000 population
PopChange	Population Change	thousands
PopGrowthRate	Population Growth Rate	percentage
DoublingTime	Population Annual Doubling Time	years

Fertility

Code	Name	Unit
Births	Births	thousands
BirthsMale / BirthsFemale	Births by sex	thousands
Births1519	Births by women aged 15 to 19	thousands
CBR	Crude Birth Rate	births per 1,000 population
TFR	Total Fertility Rate	live births per woman
NRR	Net Reproduction Rate	surviving daughters per woman
MAC	Mean Age Childbearing	years
SRB	Sex Ratio at Birth	males per 100 female births

Mortality

Code	Name	Unit
Deaths / DeathsMale / DeathsFemale	Total Deaths, by sex	thousands
CDR	Crude Death Rate	deaths per 1,000 population
LEx / LExMale / LExFemale	Life Expectancy at Birth	years
LE15, LE60, LE65, LE80 (+ Male/Female)	Life Expectancy at ages 15/60/65/80	years
InfantDeaths	Infant Deaths, under age 1	thousands
IMR	Infant Mortality Rate	infant deaths per 1,000 live births
LBsurvivingAge1	Live births surviving to Age 1	thousands
Under5Deaths	Deaths under age 5	thousands
Q5	Under-five Mortality Rate	deaths under age 5 per 1,000 live births

Code	Name	Unit
Q0040, Q0060 (+ Male/Female)	Mortality before age 40 / 60	per 1,000 live births
Q1550, Q1560 (+ Male/Female)	Mortality between ages 15–50 / 15–60	per 1,000 alive at age 15

Migration

Code	Name	Unit
NetMigrations	Net Number of Migrants	thousands
CNMR	Net Migration Rate	per 1,000 population

5. Estimates vs. projections — and the OtherVariants file (optional)

The Medium file is a single continuous series, but not all of it is equally "solid": years through the most recent observed data are **estimates** built from census and survey data, while later years are the UN's **medium-variant projection**. One plausible scenario built on stated fertility, mortality, and migration assumptions, not a certainty.

The OtherVariants file is not required, but it's there for any team that wants to go further and interrogate that uncertainty directly. It covers 2024–2101 only, and includes:

- **Alternative scenarios:** High, Low, Constant fertility, Constant mortality, Zero migration, Instant replacement, Momentum, No fertility below age 18, and several more
- **Probabilistic intervals:** Lower 80 PI, Upper 80 PI, Lower 95 PI, Upper 95 PI, Median PI, Mean — confidence-interval-style bounds from the UN's probabilistic projection model

A solid submission can be built entirely from the Medium file.

6. Suggested starting questions

- How does population age structure (median age, dependency-relevant life-expectancy indicators) diverge across countries or regions over time?
- *(Optional, uses OtherVariants)* Which countries show the widest spread between High and Low variant projections, and what does that suggest about the uncertainty in their outlook?
- *(Optional, uses OtherVariants)* How sensitive is a given country's 2100 population projection to the fertility assumption alone (compare Medium vs. Constant fertility vs. High/Low)?
- What's the relationship between net migration patterns and population growth/decline across regions?
- How might you combine several indicators into a single measure of a country's demographic profile, and what choices would that require you to justify?
- Do countries fall into distinct groups based on how their indicators have moved together over time, rather than where they stand in any single year?
- How do fertility, mortality, and migration act together to shape a country's overall population trajectory — and does that relationship differ across regions or income levels?
- Is there a single indicator in this dataset that tells the whole story on its own, or does the interesting story only emerge once you look at several together?